A GLR Optimization

Unconstrained and Constrained Maximum Likelihood Estimates (MLE)

Unconstrained MLE. Let $\hat{\theta}_t = (\hat{\theta}_{a,t}, \hat{\theta}_{b,t})$ denote the empirical rate vector at time t. The unconstrained MLE is

$$\hat{\theta}_t = \left(\frac{s_a}{n_a}, \frac{s_b}{n_b}\right) \text{ (SP)}, \qquad \hat{\theta}_t = \left(\frac{r_a}{m_a}, \frac{r_b}{m_b}\right) \text{ (EO)}, \tag{26}$$

with the convention that the test proceeds only after each group has at least one relevant observation so that the denominators are nonzero. This maximizes $L_t(\theta)$ over the parameter space $[0,1]^2$.

Constrained MLE under H_0. If the empirical disparity satisfies $|\hat{\theta}_a - \hat{\theta}_b| \leq \delta$, then the unconstrained MLE already lies in $\Theta_0(\delta)$ and therefore also maximizes the likelihood under H_0.

Otherwise, the maximizer must lie on the boundary $|\theta_a - \theta_b| = \delta$, which can be parameterized as

$$\theta_b = \theta_a + s\delta, \qquad s \in \{-1, +1\}. \tag{27}$$

Substituting this relation into the log-likelihood yields the one-dimensional function

$$f_s(\theta_a) = \ell_t(\theta_a, \theta_a + s\delta). \tag{28}$$

The feasible values of θ_a are those satisfying

$$0 \leq \theta_a \leq 1, \qquad 0 \leq \theta_a + s\delta \leq 1. \tag{29}$$

For the Bernoulli likelihood (SP case), the derivative is

$$\frac{d}{d\theta_a} f_s(\theta_a) = \frac{s_a}{\theta_a} - \frac{n_a - s_a}{1 - \theta_a} + \frac{s_b}{\theta_a + s\delta} - \frac{n_b - s_b}{1 - \theta_a - s\delta}. \tag{30}$$

An analogous expression holds for EO by replacing (n_g, s_g) with (m_g, r_g).

Since the second derivative is always negative, the Bernoulli log-likelihood is strictly concave, the function $f_s(\theta_a)$ is strictly concave over its feasible domain, and therefore admits a unique maximizer for each $s \in \{-1, +1\}$. The constrained MLE $\sup_{\theta \in \Theta_0(\delta)} \ell_t(\theta)$ is obtained by computing the maximizer for both boundary cases and selecting the solution that yields the larger likelihood value.

When the unconstrained MLE lies in $\Theta_1(\delta)$, the GLR statistic can be written as

$$\Lambda_t = \ell_t(\hat{\theta}_t) - \sup_{\theta \in \Theta_0(\delta)} \ell_t(\theta). \tag{31}$$

By symmetry, when the unconstrained MLE lies in $\Theta_0(\delta)$, the denominator is achieved at $\hat{\theta}_t$, while the numerator is given by the supremum over $\Theta_1(\delta)$. By continuity of the log-likelihood, this supremum is obtained by the same boundary optimization on $d(\theta) = \delta$ approached from the violation side, so the complementary representation is

$$\Lambda_t = \sup_{\theta \in \Theta_1(\delta)} \ell_t(\theta) - \ell_t(\hat{\theta}_t) \leq 0. \tag{32}$$

Thus, the same boundary optimization resolves both acceptance-side and rejection-side computations. The same logic applies to the Gaussian proxy audits after replacing the Bernoulli log-likelihood with the corresponding Gaussian log-likelihood.

These thresholds should therefore be interpreted as operational decision rules rather than exact finite-sample guarantees.